SIMPLE MACHINES

TIME: 1HR **MARKS: 40**

SECTION I

[***All*** *questions are compulsory*]

Question 1: **[5]**

[i] Single movable pulley is considered as a modified form of.....

[a] class I lever

[b] class II lever

[c] class III lever

[d] none of the above.

[ii] Efficiency of an ideal machine is.....

[a] Mechanical advantage / velocity ratio

[b] Mechanical advantage x velocity ratio

[c] equal to one

[d] both a and c.

[iii] Which class of lever has Mechanical advantage and velocity ratio always less than one?

[a] class III

[b] class I

[c] class II

[d] none of the above

[iv] Calculate the work done by effort when a load of 60kgf is lifted through 20m by applying an effort of 30kgf which is displaced through 60m using a block and tackle system with velocity ratio 3.

[a] 120J

[b] 1200J

[c] 180J

[d] 18000J

[v] Which type of lever is present in the diving board of swimming pool?

[a] cantilever

[b] class I

[c] class II

[d] both a and c

Question 2: **[5]**

[i] In case of a practical machine work input is …. [less / more] than work output.

[ii] In ……. [class I / class II / class III] M.A always more than one.

[iii] Single fixed pulley is a modified form of …. [class I / class II] lever.

[iv] An example of a simple machine that brings about transformation of energy is…… [sugarcane juice extractor / lever / pulley]

[v] Velocity ratio is the ratio of two ……. [dissimilar / similar] quantities hence a pure ratio and unitless quantity.

Question 3:

[a] State the class of levers and the relative position of load, fulcrum, and effort in each of the following cases:

[i] bottle opener [ii] Sugar tongs. **2**

[b] Levers of class III cannot lift heavier loads. Why? **2**

[c] A boy uses a single pulley to lift a load of 50kgf to some height. Another boy uses a single movable pulley to lift the same load to the same height. Compare the effort applied by them? **2**

d] A machine is 80% efficient. What does it mean? **2**

e] What is the M.A of class III lever and give any one application of the same. **2**

SECTION II

[*Attempt any **two** questions out of three*]

Question 4:

a] With reference to the terms mechanical advantage, velocity ratio and efficiency of a machine, name the term that will not change for a given design **3**

b] Prove that efficiency of a machine is the ratio of mechanical advantage to the velocity ratio of the machine. **4**

c] A pulley system has a velocity ratio of 4 and an efficiency of 90%. Calculate [i] Mechanical advantage of the system [ii] Effort required to raise a load of 300N by the system **3**

Question 5:

a] A pulley system has a velocity ratio equal to 4 is used to lift a load of 150 kgf through a vertical height of 20m. The effort required is 50kgf in the downward direction. Calculate:

[i] Distance moved by effort

[ii] Work done by effort

[iii] The mechanical advantage of the pulley system

[iv] The efficiency of pulley system [g = $10Nkg^{-1}$] **4**

b] Draw a neat labelled diagram of block and tackle system of pulleys having velocity ratio 4. **3**

c] State the factors on account of which the efficiency of a practical machine is always less than 100%. **3**

Question 6:

a] A cook uses a fire tong of length 28cm to lift a piece of burning coal of mass 250g. If he applies a force at 0.7m from the fulcrum find the force applied by the cook in SI unit. [g = $10ms^{-2}$] **4**

b] A crowbar of length 150cm has its fulcrum at 25 cm from the load. Calculate the Mechanical advantage of the crowbar. **3**

c] State three ways of minimizing the losses in a machine. **3**

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